Devesh Nath

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EDUCATION

PhD, Robotics - Georgia Institute of Technology Aug 2024 – May 2029 (Expected)
Third Year
BSE, Electrical Engineering - **Arizona State University** Aug 2020 – May 2024
GPA – 4.00

RESEARCH

Graduate Researcher, Trustworthy Robotics Lab – Georgia Tech Jan 2025 – Present
I use ML to discover representations that principled methods can exploit, rather than brute-forcing the problem with data, compute and empirics only. Specifically, I leverage control theory and statistical inference for provably safe world model planning, large scale conformal behavioral modelling and ultra-fast long horizon MPC.

PUBLICATIONS

Factorized Conformal Prediction Via Learned Residual-Law Retrieval

(In Submission)
Devesh Nath, Rahul Rustagi, Glen Chou

Scaling conformal prediction by learning uncertainty sufficient representations to produce tight, coverage-guaranteed uncertainty sets for multimodal behavior modelling, supporting less conservative MPC aligned with human behavior.

- Scalable, Locality-Aware Conformal Prediction
- Conformal Multimodal Autonomous Driving
- Tight, Calibrated Behavioral Modelling
- High-Dimensional Robot Reachability

Pixels to Proofs: Probabilistically Safe Latent World Model Control via Parallel Conformal Robust MPC

(RSS 2026, Robot World Model Workshop)
Devesh Nath*, Anutam Srinivasan*, Haoran Yin*, Ruitong Jiang, Jeffrey Fang, Glen Chou

Turning raw pixels into probabilistically safe robot control by combining learned latent world models, calibrated uncertainty, and GPU-parallel robust MPC.

- Pixels to Safe Control
- GPU-Parallel Robust MPC
- Conformalized World Models
- Safe Vision-Based Manipulation

Scalable Data-Driven Reachability Analysis and Control via Koopman Operators with Conformal Coverage Guarantees

(L4DC 2026, Best Paper Finalist)
Devesh Nath *, Haoran Yin *, Glen Chou

Scaling probabilistically guaranteed safety verification to high-dimensional nonlinear robotics by combining learned Koopman dynamics, efficient reachability analysis, and conformal prediction.

- Scalable Nonlinear Reachability
- Learned Koopman Linearization
- Conformal Safety Guarantees
- Black Box Dynamics

µMPC: Safe Nonlinear and World Model Control in Microseconds via Koopman-Linearized Pre-Compilation

(In Submission)
Devesh Nath, Glen Chou

Enabling microsecond scale nonlinear robotic planning. Applied for sub-millisecond vision-based planning with linear world models and fast, efficient high-dimensional robot control.

- Scalable Nonlinear Optimal Control
- Real-time control with World Models
- Fast Unitree Go2 Control
- Highly Scalable Multi-Agent Planning

Formal Safety Verification and Refinement for Generative Motion Planners via Certified Local Stabilization

(ICRA 2026)
Devesh Nath *, Haoran Yin *, Glen Chou

Bringing formal safety guarantees to diffusion, flow-matching, and vision-language motion planners through certified local stabilization, preserving multimodal behavior without retraining the generative models.

- Formally Verified Generative Planning
- Certified Local Stabilization
- Safe Multimodal Planning
- Retraining-Free Safety Refinement

INDUSTRY EXPERIENCE

Automation Engineering Intern - Rehrig Pacific Company — May 2023 – Aug 2023

- Designed and built an automated test unit for robot end effectors with application of Programmable Logic Controllers, Human Machine Interface, Industrial Sensors and Pneumatic Control Valves.
- Saved over ~ $30k per annum on Arm Tool related downtime costs.
- Programmed and deployed UR5e Cobots for repetitive pick-and-place tasks on production lines, optimizing workflows to significantly reduce manufacturing time and lower the overall cost per part produced. Used industrial sensors for part detection and synchronized UR5e for stacking using pneumatic grippers.

SKILLS

ML: World Models, Activation Steering, Generative Modeling
Controls: Nonlinear Robust Optimal Control, Reinforcement Learning, Safe Control via Statistical Reachability Analysis
Programming: Python, Jax, PyTorch, C++

LEADERSHIP

GTA – Robotics and AI Techniques — Jan 2026 – Present
Undergraduate Research Mentor – Opportunity Research Scholars — Jan 2026 – Present
Head TA - Feedback Control Systems — May 2025 – Dec 2025
President - ASU Sun Devil Robotics Club — July 2023 – May 2024
Within a semester's worth of time, I led the team to complete the mechanical design of the rover with the development of a comprehensive software stack for autonomy and control.